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ShowName: Cisco CCNA (200-301)

Topic: 2.0 Network Access

EpisodeName: Cisco Wireless Basics

Description: In this episode, you will review wireless networking principles. You'll learn why they're important to remember when setting up a wireless network.

Cisco Wireless Basics

1.11 Describe wireless principles

1.11.a Nonoverlapping Wi-Fi channels

- Channels (2.4 GHz)-3 (1, 6, 11)
- Channels (5 GHz)-24
(36,40,44,48,52,56,60,64,100,104,108,112,116,120,124,128,132,136,140,144,149,153,157,161,165)

1.11.b SSID (Service Set Identifier)

- Unique Identifier for your Wireless Network
Case sensitive
up to 32 alphanumeric characters, no spaces
- Purpose: identify, distinguish and maintain wireless connectivity

1.11.c RF (Radio Frequency)

- 2.4 GHz (802.11b,802.11g, 802.11n)
- 5 GHz (802.11a,802.11n, and 802.11ac)
- 1 - 7 GHz (802.11ax)

1.11.d Encryption

WEP (Wired Equivalent Privacy-deprecated)

- original 802.11 standard
- short encryption key (also static) with initialization vector
- used a static key encryption (doesn't change)
- could be decoded easily but many devices still support it for compatability issues

WPA

- Meant to patch the older and weaker WEP
- New system for encryption called TKIP (Temporal Key Integrity Protocol)
- now considered susceptible to wireless attacks
- should not be used in Enterprises

WPA2 (PSK--Pre Shared Key)

- currently most secure using AES (Advanced Encryption Standard)
- PSK (Pre-Shared-key) to authenticate
- PSK means static key configured on client and AP

WPA2 (Enterprise)

Currently, most secure using AES and 802.1x for authentication (RADIUS, PEAP/MSCHAPv2 etc.)